

Where Should the Long Wires Go? Total-Wire, Longest-Wire and Performance Trade-offs in Radix-4 On-Chip Networks

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Abstract

A 2D mesh leaves $4n$ router ports unused on the boundary of an $n \times n$ grid. Tori spend them on wraparound links; other designs fold links back along the edges to save wire. Which choice is best depends on two different wire costs: total wire, which sets area and link energy, and the longest wire, which limits the clock. We formulate the use of these spare ports as a matching problem that contains the mesh, the torus and edge-folded designs, and search it under explicit total-wire and longest-wire constraints. Designs are evaluated with a cycle-level simulator validated against BookSim 2, seven dependency-driven PARSEC traces, and DSENT area, energy and timing models at 22 nm. On an 8×8 network, a searched layout whose extra links are at most five tiles long is the fastest evaluated design at a fixed 2 GHz clock, with 17% lower runtime and 23% lower energy-delay product than a mesh under compressed traffic, using 20% less wire than a torus. When the longest wire limits the clock, the folded torus is best, and edge-folded designs that keep full-length wraparounds lose to the mesh.

Keywords: network-on-chip, topology, wire length, deadlock-free routing, simulated annealing, reinforcement learning, DSENT, BookSim

1. Introduction

Most many-core chips still connect their tiles with a 2D mesh. It maps directly onto the floorplan, every link is one tile long, and every router has the same five ports (four neighbours and the local core). Its weakness is distance: a packet may need $2(n - 1)$ hops on an $n \times n$ grid. The classic

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6 remedy is to add long links. A torus closes every row and column into a
 7 ring; express links, flattened butterflies and low-diameter graphs go further
 8 [1, 2, 3].

9 Long links are paid for twice. Their *total* length sets wire area, repeater
 10 area and link energy. Their *maximum* length sets how fast the network can be
 11 clocked, because a wire that does not fit in one cycle either lowers the clock or
 12 must be pipelined. Recent chiplet work makes the second cost explicit: Kite
 13 classifies interposer topologies by their longest link and shows that long-link
 14 topologies lose much of their hop advantage once each runs at the clock its
 15 longest wire allows [4]. NetSmith automates the same design problem with
 16 mixed-integer programming under link-length limits [5].

17 This paper studies the on-chip, radix-4 version of the question. A mesh
 18 router on the boundary of the grid has one or two unused ports, $4n$ in total.
 19 Pairing these spare ports adds $2n$ links without changing the router, and
 20 every such pairing is a valid radix-4 network. The torus is one pairing; edge-
 21 folded designs that link boundary routers along the same edge are others.
 22 The question is simple to state: *given that wires cost area, energy and clock*
 23 *speed, where should the long links go?*

24 We make the following contributions.

- 25 • A formulation of spare-boundary-port use as a perfect-matching design
 26 space, with optional constraints on total wire and on the longest link,
 27 that contains the mesh, the torus and edge-folded designs as special
 28 cases (Section 3).
- 29 • An evaluation methodology combining a cycle-level simulator validated
 30 against BookSim 2, dependency-driven PARSEC traces, and DSENT
 31 area, energy and timing models, under two clocking regimes (Sections 4
 32 and 5.1).
- 33 • Quantitative trade-offs for six designs on an 8×8 network (Sections 5.2–
 34 5.5). The main findings are that a searched design with links of at most
 35 five tiles is the best performer among the evaluated designs at a fixed
 36 clock, that the folded torus is best when the longest wire sets the clock,
 37 and that saving total wire while keeping full-length links, as edge-folded
 38 designs do, buys area but not speed.

39 One member of this design space, an edge-folded “modified torus”, was pro-
 40 posed in earlier undergraduate work [6] and re-evaluated in a preprint [7];
 41 here it is one design point among many. We also compare simulated anneal-
 42 ing with REINFORCE and random search as search methods (Section 5.6).
 43 All code, traces-processing tools, raw results and designs are open source (see
 44 Data Availability).

45 2. Background and related work

46 Table 1 summarises the work closest to ours. Two papers are particularly
 47 relevant. *Kite* [4] designs network-on-interposer topologies for a 64-core, four-
 48 chiplet system with 20 concentrated routers of radix up to 8. It introduces
 49 an effective hop count, the average hop count divided by the clock frequency
 50 the longest link permits, and shows with DSENT that topologies with long
 51 links run at lower clocks (2.7 GHz against 4.0 GHz for a mesh at 22 nm).
 52 *NetSmith* [5] generates interposer topologies with a mixed-integer program
 53 that minimises average hop count or maximises sparsest-cut bandwidth under
 54 router-radix and link-length limits, and reports that solve times grow from
 55 minutes for 20 routers to about two days for 48. In its evaluation, the expert-
 56 designed Kite-Small turned out to be optimal for one objective.

57 Our setting differs from both in three ways. The network is a monolithic
 58 2D NoC with one router per tile and radix 4. The design space is restricted
 59 to pairings of the mesh’s spare boundary ports, which keeps the mesh intact
 60 and is therefore compatible with existing tiled floorplans. And total wire is
 61 treated as a budget to be traded against performance, alongside the longest-
 62 link limit, rather than only as a consequence of the topology.

63 Deadlock freedom follows Duato’s theory [16]: adaptive routing is deadlock-
 64 free if an escape sub-network with an acyclic channel dependency graph [36]
 65 is always reachable. Up*/down* routing [17] provides such a sub-network on
 66 any connected graph, which matters because searched designs are irregular.
 67 Kite and NetSmith likewise use shortest-path routing with escape VCs.

68 3. Design space

69 Number the routers of an $n \times n$ grid by row and column. In a mesh, each
 70 corner router has two unused ports and each other boundary router has one,
 71 $4n$ ports in total. A *design* is a perfect matching of these ports such that no
 72 link is a self-loop, duplicates a mesh link, or duplicates another added link.
 73 Every design keeps all mesh links, adds $2n$ links, and leaves every router with
 74 radix 4.

- 75 • The **torus** pairs every port with the port on the opposite edge.
- 76 • The **modified torus** [6] keeps the corner wraparounds (and the middle
 77 row and column for odd n) and links every other boundary router to
 78 the router $\lfloor (n - 1)/2 \rfloor$ positions away on the same edge, shortening
 79 most extra links.

- The **folded torus** [1] is not in the space, because it replaces mesh links. It is the torus with each ring folded so that no link is longer than two tiles, the standard practical torus layout and a baseline in Kite and NetSmith. We model it as the torus graph with the *same core-to-router mapping* and only the physical link lengths changed, so that it receives exactly the torus’s traffic and differs only in wire length, clock and energy.

Two constraints can be imposed: a total-wire budget B (sum of Manhattan link lengths, in tile pitches) and a longest-link limit $L_{\max}$. Each design is scored by

$$J(G) = \frac{1}{|P|} \sum_{p \in P} \frac{\hat{\Theta}_p(G)}{\hat{\Theta}_p(\text{torus})} - 0.25 \frac{\bar{h}(G)}{\bar{h}(\text{torus})} - 5 \frac{\max(0, W(G) - B)}{B}, \quad (1)$$

where $\hat{\Theta}_p$ is the analytical throughput estimate (Section 4.1) under traffic pattern $p \in P = \{\text{uniform, transpose, bit-complement, tornado}\}$, $\bar{h}$ is the mean hop count and W the total wire. The longest-link limit is enforced by restricting the space. Averaging over several patterns matters: an earlier version that optimised uniform traffic alone produced designs that looked better analytically but were worse in simulation on transpose traffic [7].

4. Methodology

4.1. Analytical metrics

Hop distances and shortest-path counts come from breadth-first search. With each flow split evenly over its minimal paths, the load on directed channel $c = (u \rightarrow v)$ per unit injection rate is $\gamma_c = \sum_{s,t} T_{st} \sigma_{su} \sigma_{vt} / \sigma_{st} \mathbf{1}[d(s,u) + 1 + d(v,t) = d(s,t)]$, where T is the traffic matrix, σ counts shortest paths and d is hop distance. The throughput estimate is $\hat{\Theta} = 1 / \max(\max_c \gamma_c, \max_t \sum_s T_{st})$, the second term being the ejection limit. It reproduces the textbook torus bound [1] and is used only as a fast search proxy; every conclusion is checked in simulation.

4.2. Routing and deadlock

All designs use minimal adaptive routing on VCs $1..V-1$ with an up*/down* escape channel on VC 0, rooted at the most central router. A blocked packet may enter the escape channel at any hop and may return to an adaptive

107 VC later. Because flow control is virtual cut-through, Duato’s condition for
 108 cut-through networks applies [37]: routing is deadlock-free if the escape rout-
 109 ing subfunction is connected and its channel dependency graph, with only
 110 *direct* dependencies, is acyclic. Up*/down* is connected on any connected
 111 graph, and from every router and up*/down* phase it offers a legal next hop
 112 toward every destination, so a packet in any buffer can always request an
 113 escape channel. We build the escape channel dependency graph for every
 114 design, including all legal phase transitions and fresh entries into the escape
 115 channel, and check it for cycles. As an empirical check we also rerun the ap-
 116 plication traces with packets confined to the escape channel once they enter
 117 it (Section 5.5). Shortest-path routing on a single VC, as used in [6], has a
 118 cyclic dependency graph on the torus and modified torus and deadlocks in
 119 simulation in every run at injection rates of 0.2 and above; the escape scheme
 120 never deadlocks, including at loads far beyond saturation.

121 4.3. Cycle-level simulator

122 The simulator models input-queued routers with $V = 3$ VCs, credit-based
 123 flow control and virtual cut-through: a packet of s flits advances only if the
 124 downstream VC has s free slots, then holds its input port and output link
 125 for s cycles. Each port carries one flit per cycle. A hop costs one router
 126 cycle plus the link delay. A 2-cycle minimum gap between packets leaving
 127 the same VC models the allocation pipeline of BookSim’s router; Section 5.1
 128 shows that this reproduces BookSim’s throughput. Synthetic runs use open-
 129 loop Bernoulli injection with warm-up, a tagged measurement window and
 130 full drain.

131 4.4. Application traces

132 We replay Netrace traces [34] of seven PARSEC benchmarks [38] (blacksc-
 133 holes, canneal, ferret, fluidanimate, swaptions, vips, x264) captured on a 64-
 134 core 8×8 CMP. For each, the first 250 000 packets of the region of interest are
 135 replayed with their recorded dependencies: a packet is injected only after the
 136 packets it depends on have arrived, plus its recorded compute gap minus the
 137 original network’s zero-load latency. Completion time is therefore a runtime
 138 proxy. Control packets are 1 flit and data packets 5 flits (16 B flits). Because
 139 the traces are light at their recorded speed (about 10^{-3} packets/node/cycle),
 140 we also compress compute gaps by $10\times$ and $50\times$, emulating faster cores. The
 141 cores ran at 2 GHz, and gaps are converted to real time with that clock so
 142 that networks running at different frequencies are compared fairly.

143 4.5. Area, energy and timing

144 We characterise routers (128-bit flits, 3 VCs $\times$ 8-flit buffers) with 3, 4
145 and 5 ports and repeated global-layer links with DSENT [35] at 22 nm.
146 Each router is modelled with its actual port count: a mesh’s corner and edge
147 routers have 3 and 4 ports, every router of the radix-4 designs 5 (a 3-port
148 router has 58% of the leakage and 57% of the area of a 5-port one). For
149 the 5-port router, dynamic energy is 3.63 pJ per flit (buffer write and read,
150 crossbar, arbitration), leakage 30.4 mW and area 0.037 mm². Link energy
151 is 3.10 pJ per flit per mm. With repeaters sized to meet one clock period,
152 the longest wire DSENT can fit in a cycle is 13.0 mm at 2 GHz, 8.5 mm
153 at 3 GHz and 6.5 mm at 4 GHz; the router meets 4 GHz. Link area is the
154 global-wire track area of two 128-bit channels plus repeaters. Network energy
155 is router and link dynamic energy from simulated activity, clock-tree energy
156 per router per cycle, and leakage over the run.

157 Two clocking regimes are evaluated, both with 1.5 mm tiles unless stated:

- 158 • **Fixed clock** (2 GHz): every network runs at 2 GHz. All links up to
159 8.7 tiles fit in one cycle; longer links are pipelined.
- 160 • **Wire-limited clock** (as in Kite and NetSmith): each network runs at
161 the highest frequency at which its longest link fits in one cycle, capped
162 at 4 GHz.

163 5. Results

164 5.1. Simulator validation against BookSim

165 BookSim cannot run the escape-channel routing, so the validation uses
166 dimension-order routing, which both simulators implement identically, in-
167 cluding BookSim’s dateline VC rule for tori. Router delays are matched (4
168 cycles per hop), as are buffer sizes and packet lengths; BookSim’s 6-cycle
169 injection/ejection overhead and its self-addressed packets are corrected for
170 explicitly. Across the mesh, the 4-VC torus and 5-flit packets (twelve configu-
171 rations), zero-load latency agrees within 2.1%, the mean latency error below
172 saturation is 0.2–5.3%, and saturation points differ by at most 0.06 pack-
173 ets/node/cycle (Fig. 1). On the 2-VC torus, where each dateline class has
174 a single VC, BookSim saturates much earlier (0.30 against 0.42 on uniform
175 traffic) because a single VC cannot forward back-to-back packets through its
176 allocation pipeline. Adding a 2-cycle VC turnaround to our router repro-
177 duces BookSim’s saturation points exactly on all three patterns (0.32, 0.12
178 and 0.20 against 0.30, 0.12 and 0.20). All remaining experiments use this
179 turnaround.

5.2. Static trade-offs

Table 2 and Fig. 2 introduce the designs. The two searched designs come from the two constraint types: the best design found with a total-wire budget of 208 tiles, and the best found with no extra link longer than five tiles.

Fig. 3 shows the two frontiers. Under a total-wire budget (left), throughput rises slowly from 152 to 208 tiles (0.52 to 0.71 of the torus) and then jumps to the torus itself at 224 tiles. The modified torus lies on this frontier: at its 176-tile budget no method found a better design. Under a longest-link limit (right), the best designs reach 0.52, 0.60 and 0.65 of the torus at limits of 2, 3 and 5 tiles. The folded torus, outside the space, dominates all of them: the same longest link as the 2-tile designs and the torus’s throughput (it is the torus graph with the same core mapping, so its analytical metrics equal the torus’s). Its cost is wire: 224 tiles against 144–180 for the searched designs. The 5-tile design improves on the modified torus in hop count (3.84 against 4.13), throughput and longest link (5 against 7 tiles) for 2% more wire.

5.3. Area and timing

Wire dominates network area: at 1.5 mm tiles, global-wire tracks are 92% of the torus’s 30.0 mm². The modified torus and the 5-tile design save 20% and 18% of the torus’s area. Under a wire-limited clock, the torus and modified torus, whose 10.5 mm wraparounds fit in a cycle only up to 2.45 GHz, are slower than the mesh at zero load (3.32 and 3.37 ns, against 2.67 ns), and the 9-tile searched design is slower still (4.08 ns). The folded torus runs at 4 GHz and is fastest (2.03 ns); the 5-tile design follows (2.25 ns at 3.42 GHz). At a fixed 2 GHz clock the two searched designs have the lowest zero-load latencies (3.84 and 3.86 ns, against 4.06 ns for the torus and folded torus). Fig. 4 shows that tile pitch shifts these rankings: at 1 mm the 5-tile design runs at 4 GHz and becomes the fastest network under a wire-limited clock (1.92 ns), while at 2 mm the wire-limited clocks of the long-link designs fall to 1.75 GHz (torus, modified torus) and 2.57 GHz (5-tile).

5.4. Synthetic traffic

Fig. 5 shows load–latency curves under four synthetic patterns. The searched designs have the lowest zero-load latency on uniform and transpose traffic (7.65 and 6.85 cycles for the 5-tile design at 2 GHz, against 8.06 and 9.06 for the torus), consistent with their lower hop counts. Their saturation throughput, however, is not uniformly good. On uniform traffic the torus and folded torus saturate highest (0.45 packets/node/cycle, against 0.38 for the 5-tile design and the mesh); on transpose the 5-tile design saturates at 0.20, below the mesh (0.31), because several transpose flows share its few

219 long links. At 2 GHz the folded torus behaves exactly like the torus, as
 220 it should: same graph, same traffic, and every link fits in one cycle. The
 221 two tori saturate highest on uniform, transpose and bit-complement traffic;
 222 on tornado traffic the modified torus (0.30) and the 5-tile design (0.28) do
 223 better than both tori (0.25). The modified torus is the weakest long-link
 224 design on bit-complement traffic, saturating at 0.10, below the mesh. At
 225 4 GHz the two-cycle wraparounds raise the zero-load latency of the torus
 226 and modified torus by 0.2–0.9 cycles, while the folded torus keeps one-cycle
 227 links. The application traces run far below these saturation points even
 228 at 50 $\times$ compression, which is why the searched designs’ latency advantage,
 229 rather than their throughput, determines the application results below.

230 5.5. Applications and energy

231 All 252 trace runs completed without deadlock. At the recorded speed
 232 (1 $\times$), every design finishes within 1.1% of the mesh: at the load of these traces
 233 the network is rarely on the critical path, and because every radix-4 design
 234 has more router ports than the mesh’s edge routers and more link repeaters,
 235 all of them consume 11–17% *more* network energy than the mesh. The
 236 comparison of the long-link designs becomes meaningful under compression
 237 (Table 3, Fig. 6). Table 4 breaks the 10 $\times$ results down by benchmark: the
 238 rankings below hold on six of the seven benchmarks, and the seventh (vips)
 239 is insensitive to the network, with every design within 0.1% of the mesh.

240 *At a fixed clock*, the 5-tile searched design is the best evaluated network
 241 at both compression levels: at 50 $\times$ it runs 10% faster than the torus and
 242 folded torus (which behave identically at this clock) and 9% faster than
 243 the modified torus, and has the lowest energy-delay product (0.608 of the
 244 mesh). Network energy is dominated by leakage and clock power, so it follows
 245 runtime; dynamic energy alone is lowest on the mesh and 2%, 4%, 7% and 8%
 246 higher on the 5-tile design, modified torus, folded torus and torus respectively.

247 *With a wire-limited clock*, the folded torus is the best evaluated design
 248 (runtime 0.827 and EDP 0.771 of the mesh at 50 $\times$). The torus and modified
 249 torus, held to 2.45 GHz by their wraparounds, run 23–25% *slower* than the
 250 mesh, and the 9-tile searched design is worse still. The 5-tile design is second,
 251 1.5% behind the folded torus at 10 $\times$ and 3% at 50 $\times$, with 18% less area.

252 *Tile pitch*. Repeating the 10 $\times$ runs at 1.0 and 2.0 mm tiles confirms the
 253 static picture. At 1.0 mm the 5-tile design is the best or tied-best evaluated
 254 design in both regimes (wire-limited runtime 0.870 and EDP 0.847, against
 255 0.909 and 0.925 for the folded torus). At 2.0 mm, under a wire-limited clock,
 256 the folded torus is the only design that beats the mesh, and the 5-tile design
 257 falls behind it (runtime 1.035). At a fixed clock the 5-tile design has the

lowest runtime at every pitch (tied within 0.1% with the 208-tile design at 1.0 mm).

Escape-channel rule. Rerunning the $10\times$ and $50\times$ experiments with packets confined to the escape channel once they enter it changes runtime and energy by at most 0.2% for every design and regime, so the results do not depend on letting packets return to adaptive channels.

5.6. Search methods (secondary)

Table 5 compares the search methods at equal evaluation budgets. Annealing from random starts comes within 0.03 of the modified torus; annealing started from it finds nothing better, so the hand design is (empirically) locally optimal at its budget, much as NetSmith found Kite-Small optimal for one of its objectives [5]. REINFORCE, with a tabular autoregressive policy that picks a partner for each port, lies between random search and annealing. Its policy has a separate logit for every port pair and no notion of geometry, so it cannot generalise from one pairing to a similar one; a policy that shares parameters across positions, such as a graph neural network, is the natural next step, and learned performance models [28, 31] could replace the analytical proxy inside the search. Each annealing run takes a few minutes for 64 routers, against the multi-day MILP solves NetSmith reports for 48 [5], at the price of no optimality guarantee.

6. Discussion

Which design to use. The results separate the two wire costs. If the clock is set by the longest wire, length is what matters and, among the evaluated designs, the folded torus wins: it has the torus’s graph, links of at most two tiles, and the best performance at 1.5–2 mm pitches, at the cost of the most wire. If links can be pipelined or the clock is set elsewhere, a searched design that keeps links moderately long (at most five tiles) and saves 18% of the torus’s area is the fastest evaluated network. Designs that save total wire but keep a few full-length links, including the modified torus, give up most of the torus’s advantage in exchange for area: they are never the best choice on performance.

Threats to validity.

- *Traces.* Netrace’s traces come from in-order cores and are light; the network barely affects runtime at recorded speed. Compression emulates faster cores but not a different memory system. Full-system simulation, as in NetSmith, would be stronger.

- 294 • *Physical model.* Areas and energies come from DSENT models, not
295 layout; wire length is the Manhattan distance between tiles (or the
296 folded layout’s length for the folded torus); DSENT’s 22, 32 and 45 nm
297 models share global-wire dimensions.
- 298 • *Simulator.* Virtual cut-through rather than wormhole and a greedy
299 separable allocator. The validation against BookSim covers dimension-
300 order routing only, as BookSim cannot run the escape scheme.
- 301 • *Scope.* One grid size (8×8), radix 4, boundary-port designs only; the
302 folded torus shows that rewiring interior links can do better, which
303 this space excludes by construction. We do not compare against a
304 topology generated by NetSmith or the Library of Networks: both
305 target concentrated interposer networks with a different router layout
306 and radix, NetSmith relies on a commercial MILP solver, and at 64
307 routers its reported solve times run to days; a like-for-like comparison
308 would need its formulation re-targeted to monolithic radix-4 grids.
- 309 • *Folded-torus mapping.* If cores instead keep their tile positions and only
310 the wiring is folded, physically defined permutation patterns map onto
311 different flows and the folded torus’s analytical throughput rises to 1.26
312 of the flat torus; we use the same-mapping comparison throughout.

313 7. Conclusion

314 Pairing a mesh’s spare boundary ports is a small, floorplan-compatible
315 design space that already contains the torus and edge-folded designs. Our
316 evaluation, grounded in a BookSim-validated simulator, PARSEC traces and
317 DSENT, shows that the right design depends on which wire cost dominates.
318 When the longest wire sets the clock, the folded torus is the best of the
319 evaluated designs. At a fixed clock, a searched design whose extra links are
320 at most five tiles long reduces runtime by 17% and energy-delay product
321 by 23% against a mesh under compressed traffic, outperforms the torus and
322 folded torus, and needs 20% less wire than a torus. Saving total wire while
323 keeping full-length links, the idea behind the design this study started from,
324 buys area but not performance.

325 Data availability

326 The simulator, analysis code, trace-conversion and DSENT drivers, searched
327 designs and all raw results are available at [https://github.com/R4hulR/](https://github.com/R4hulR/nocsim)
328 `nocsim`. The Netrace traces are available from their authors [34].

329 CRediT authorship contribution statement

330 **Rahul Ray:** Conceptualization, Methodology, Software, Validation, For-
331 mal analysis, Investigation, Data curation, Writing – original draft, Writing
332 – review & editing, Visualization.

333 Declaration of competing interest

334 The author declares that they have no known competing financial inter-
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Table 1: Related work and the gap this paper addresses.

Theme	Representative work	Gap
Low-diameter on-chip topologies	Slim NoC [2], Sparse Hamming Graph [3]	Lower diameter through routers with more ports; we keep the radix fixed at 4 and ask how to spend wire.
Length-constrained topologies	Butter Donut [8], Kite [4], NetSmith [5], Library of Networks [9]	Chiplet interposers with concentration and radix up to 8; link length as a hard limit. We study a monolithic radix-4 NoC and treat total wire as a budget.
Physical-design-aware topology	HexaMesh [10], PlaceIT [11], FoldedHexaTorus [12], wafer-scale co-design [13, 14], Rapid-Chiplet [15]	Placement and packaging co-designed with topology; we bring wire-length and DSENT energy to a 2D on-chip network.
Deadlock freedom on irregular graphs	Duato [16], up*/down* [17], Static Bubble [18], SPIN [19], BINDU [20], SWAP [21], modular chiplet routing [22, 23, 24]	Recovery schemes avoid VC overhead; we use avoidance with an escape VC so every searched design is provably deadlock-free.
ML for NoCs	RL routing [25, 26, 27], learned performance models [28, 29, 30, 31]	Mostly routing decisions or fast predictors; we apply RL to topology search and compare it with annealing.
Evaluation tools	BookSim [32], Garnet [33], Netrace [34], DSENT [35]	Topology studies often use synthetic traffic alone; we combine validated simulation, application traces and physical models.

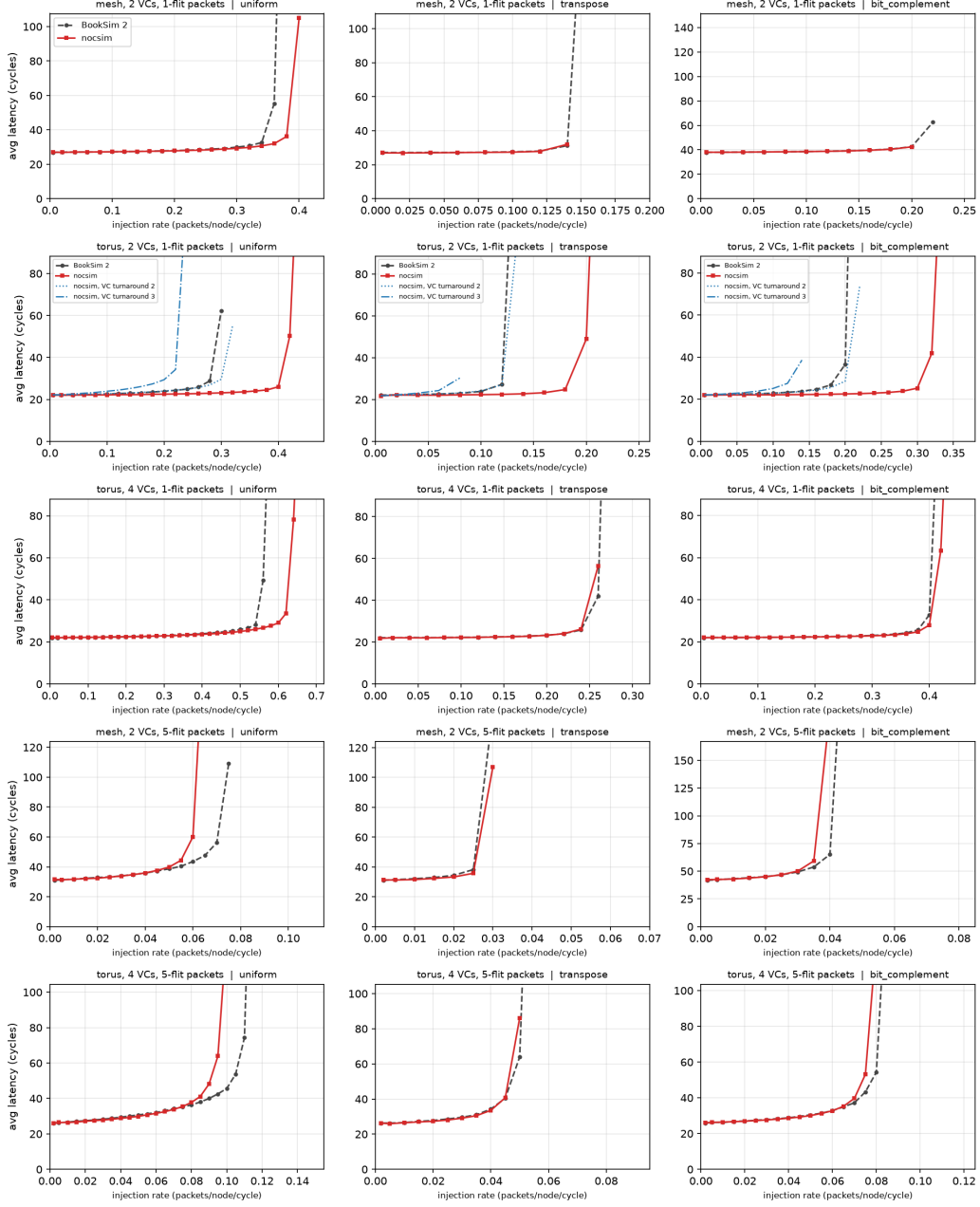


Figure 1: Our simulator (red) against BookSim 2 (grey, dashed): dimension-order routing on 8×8 networks, three seeds per point. Rows: mesh with 2 VCs, torus with 2 VCs (one VC per dateline class, with additional curves for a 2- and 3-cycle VC turnaround), torus with 4 VCs, and 5-flit packets on the mesh and 4-VC torus.

Table 2: The six designs at 8×8 (1.5 mm tiles, 22 nm). Wire and longest link in tile pitches; throughput estimate averaged over four patterns, relative to the torus; area and wire-limited clock from DSENT.

Design	Wire	Longest	Hops	Thr. est.	Area (mm ²)	Max clock (GHz)
Mesh	112	1	5.33	0.56	15.9	4.00
Torus	224	7	4.06	1.00	30.0	2.45
Folded torus	224	2	4.06	1.00	29.9	4.00
Modified torus	176	7	4.13	0.63	24.1	2.45
Searched (links ≤ 5)	180	5	3.84	0.65	24.5	3.42
Searched (wire ≤ 208)	208	9	3.79	0.71	28.0	1.86

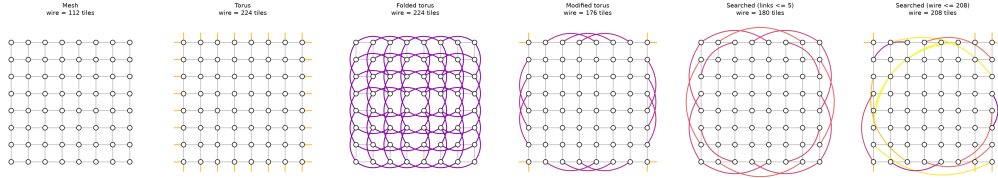


Figure 2: The six designs. Grey: mesh links. Edge stubs: full-length wraparounds. Arcs: other added links, coloured by length.

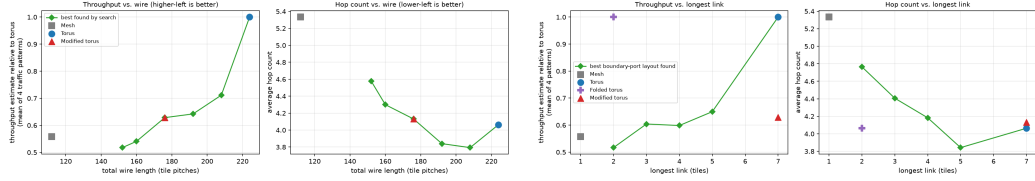


Figure 3: Left: best design found for each total-wire budget. Right: best design found for each longest-link limit, against the reference designs. Both: throughput estimate relative to the torus (mean of four patterns).

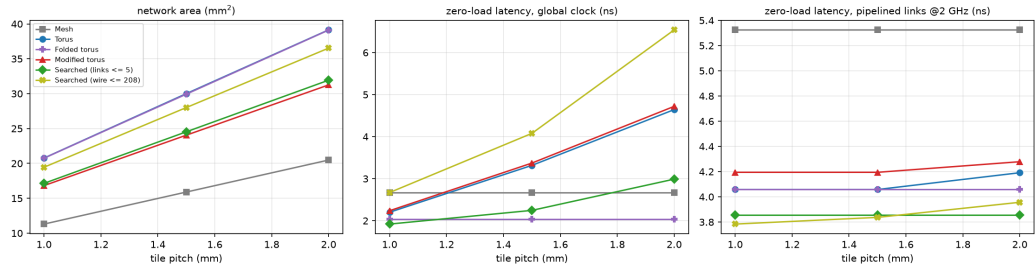


Figure 4: Area and zero-load latency against tile pitch (22 nm, DSENT). Middle: wire-limited clock. Right: fixed 2 GHz clock.

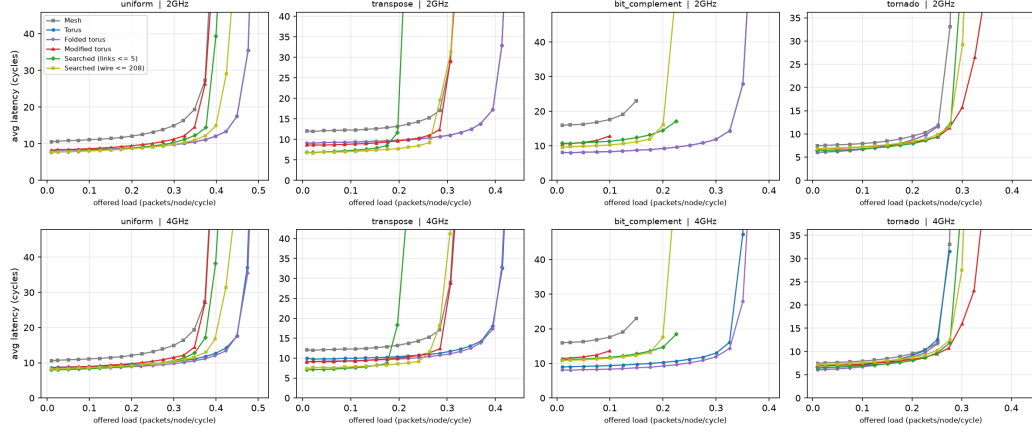


Figure 5: Synthetic traffic: average latency (cycles of each clock) against offered load. Top row: 2 GHz (all links of the hand designs take one cycle). Bottom row: 4 GHz (one cycle reaches 4.3 tiles, so the 7-tile wraparounds take two).

Table 3: PARSEC traces (seven benchmarks, geometric mean), relative to the mesh; lower is better. EDP: network energy $\times$ runtime.

Design	Fixed 2 GHz clock				Wire-limited clock			
	10 $\times$		50 $\times$		10 $\times$		50 $\times$	
	Run	EDP	Run	EDP	Run	EDP	Run	EDP
Torus	0.878	0.908	0.812	0.778	1.115	1.435	1.246	1.781
Folded torus	0.878	0.864	0.812	0.741	0.909	0.926	0.827	0.771
Modified torus	0.872	0.863	0.803	0.734	1.108	1.366	1.233	1.682
Searched (links ≤ 5)	0.826	0.770	0.733	0.608	0.923	0.955	0.852	0.816
Searched (wire ≤ 208)	0.829	0.796	0.738	0.633	1.202	1.628	1.425	2.269

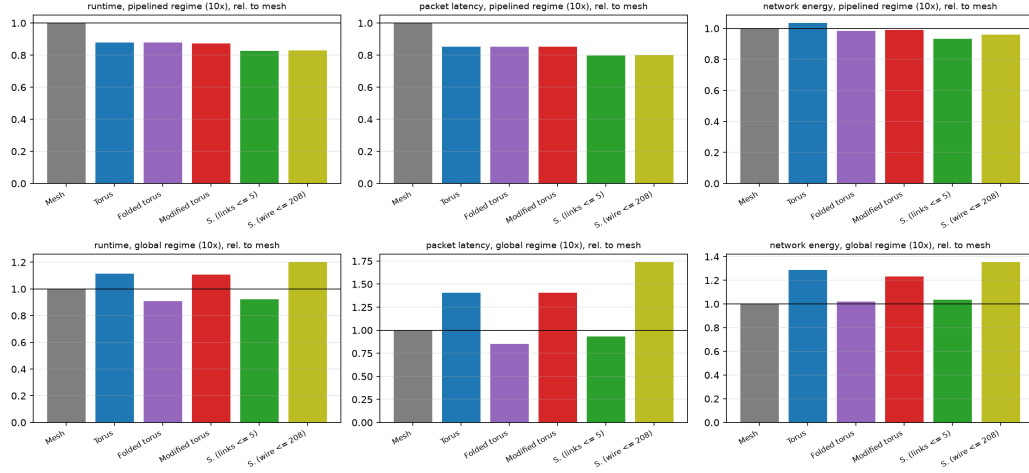


Figure 6: PARSEC traces at 10 $\times$ compression, relative to the mesh (geometric mean of seven benchmarks). Top: fixed 2 GHz clock. Bottom: wire-limited clock.

Table 4: Runtime relative to the mesh for each PARSEC benchmark (10 $\times$ compression, 1.5 mm tiles); lower is better. Best evaluated design per row in bold.

Benchmark	Fixed 2 GHz clock					Wire-limited clock				
	Torus	Folded	Modified	S. ≤ 5	S. ≤ 208	Torus	Folded	Modified	S. ≤ 5	S. ≤ 208
blackscholes	0.811	0.811	0.801	0.745	0.749	1.139	0.848	1.125	0.867	1.2
canneal	0.859	0.859	0.854	0.806	0.808	1.134	0.896	1.129	0.918	1.2
ferret	0.895	0.895	0.891	0.844	0.849	1.114	0.928	1.110	0.941	1.1
fluidanimate	0.878	0.878	0.871	0.822	0.824	1.118	0.914	1.110	0.926	1.1
swaptions	0.848	0.848	0.851	0.795	0.800	1.146	0.883	1.150	0.911	1.2
vips	0.999	0.999	0.999	0.999	0.999	1.000	1.000	1.000	1.000	1.0
x264	0.866	0.866	0.848	0.790	0.794	1.160	0.899	1.137	0.905	1.2
Geo. mean	0.878	0.878	0.872	0.826	0.829	1.115	0.909	1.108	0.923	1.2

Table 5: Search at the modified torus’s 176-tile budget: 2500 objective evaluations, three seeds. Throughput relative to the torus.

Method	J (mean $\pm$ std)	Thr.	Hops
Random search	-0.03 ± 0.08	0.42	3.92
REINFORCE	0.25 ± 0.03	0.50	4.10
Simulated annealing	0.34 ± 0.02	0.59	4.04
Modified torus	0.37	0.63	4.13